Business Requirements Document (BRD)

AI Powered Media Content Moderation & Reporting System

Methodology reference: Kendall & Kendall, *Systems Engineering Analysis and Design*, 11th Ed.

Abstract

This document presents the business requirements and feasibility analysis for an AI-powered continuous content detection and reporting system designed for the [confidential]. It operates across ~14 live television channels and five social media platforms (Instagram, TikTok, YouTube, Facebook, and Reddit). Rather than performing content takedown, the system's scope is limited to detection, classification, and structured reporting of potentially harmful/non-compliant material. The proposed architecture applies systems analysis principles to decompose the problem into three parallel processing layers: (1) audio (speech-to-text), (2) text (on-screen OCR and native social text), and (3) video/image classification, each supporting both Urdu and English, including Roman-Urdu and code-switched speech.

Three implementation strategies are evaluated using a feasibility framework (Kendall & Kendall, 2023) spanning technical, economic, operational, and schedule dimensions: (1) a full-scale, commercially-hosted implementation using managed cloud APIs, estimated at $7,000–$22,500 per month in technology cost ofr a fully custom, (2) self-hosted implementation built entirely in-house on open-source AI models, estimated at $500–$1,250 per month in technology cost, plus a one-time $3,000–$5,000 hardware investment. (3) a recommended middle-ground implementation (Moderate Tier) that self-hosts on in-house owned GPU hardware while relying on pre-built, production-ready open-source deployment software to minimize engineering effort aand is estimated at $1,300–$2,800 per month in technology cost, plus a one-time $5,000–$7,500 hardware investment. The analysis finds that the Moderate Tier the most feasible option overall, capturing most of the cost savings of self-hosting while avoiding the multi-month custom-engineering timeline of a fully custom-engineered system.

1. Problem Statement

The organization has no continuous, systematic way to detect non-compliant or policy-violating content across its two digital media streams. Content moves through three distinct forms: (1) spoken audio (live TV commentary, video posts), (2) on-screen or written text (tickers, captions, social posts, comments), and (3) video imagery, in both Urdu and English, often mixed within the same broadcast or post. Manual review alone cannot keep pace with approximately 14 channels running 24/7 plus continuous social posting, resulting in inadequate reporting and reputational risk. No audit trail currently exists to demonstrate reporting history or escalation evidence.

2. Proposed Solution

The proposed system is a continuous, AI-driven detection and reporting pipeline that monitors both streams and classifies content by severity, without taking any enforcement action itself (see Section 3, Scope). It is built around three parallel processing layers.

Audio layer (speech-to-text)

All ~14 TV channels' audio, puls audio tracks from video posts on YouTube/Instagram/Facebook/TikTok, are transcribed in near-real time using a speech-recognition model that natively support both Urdu and English, including code-switched speech.

Text layer (written/on-screen content)

On-screen text (tickers, captions, graphics) is extracted via OCR tuned for Urdu (Nastaliq) and Latin script along with native text (social posts and comments) is pulled directly via each platform's API. Both are run through a bilingual keyword/regex engine and a machine-learning classifier trained on Urdu, English, and Roman-Urdu.

Video/image layer

Visual frames from TV and social video are periodically sampled and passed through an image classifier for visual policy violations.

All three layers write to a single flagged-item queue, tagged by severity. A small human team reviews only the highest-severity flags, refines keyword lists and models, and compiles the audit trail into formatted reports.

3. Project Scope

In scope:

- Continuous monitoring of ~14 live TV feeds (audio, on-screen text, video) for potential compliance or editorial violations.
- Continuous monitoring of content on Instagram, TikTok, YouTube, Facebook, and Reddit (text, audio-in-video, and image).
- AI-based detection across audio, text, and video layers, in Urdu and English.
- Human review, classification, and severity-tagging of AI-flagged items only.
- Structured reporting to internal Compliance/Legal teams.
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Out of scope:

- Content takedown, blocking, muting, or removal of any kind, on any platform.
- Direct interaction with regulatory agencies, enforcement actions, or legal proceedings.
- Automated or manual transmission interruption for live TV.
- Moderation of platforms other than the five named social networks.
- Moderation of pre-recorded/archival content not tied to the live broadcast schedule.

4. Business Objectives

1. Achieve continuous (24/7/365) detection coverage across all ~14 TV channels and five social platforms, for audio, text, and video content.
2. Support both Urdu and English (including Roman-Urdu and code-switched speech).
3. Reduce time-to-awareness for potentially non-compliant content.
4. Produce accurate, audit-ready reports for internal compliance and reporting obligations.
5. Deliver the solution at the lowest sustainable cost and shortest feasible development timeline, using no more than 5–6 human supervisors.
6.

5. Functional Requirements

- FR1: System shall ingest live audio/video feeds from all ~14 TV channels with time-stamped logging.
- FR2: System shall transcribe audio in Urdu and English (including code-switched speech) in near-real time.
- FR3: System shall extract on-screen text (tickers, captions, graphics) via OCR supporting Urdu script.
- FR4: System shall connect to Instagram, TikTok, YouTube, Facebook, and Reddit APIs for read-only content and comment monitoring.
- FR5: System shall classify text (native, transcribed, and OCR-derived) for policy violations, including Roman-Urdu.
- FR6: System shall sample and classify video frames for visual policy violations.
- FR7: System shall route flagged items to a human review queue with severity tiers, for classification only.
- FR8: System shall maintain an immutable audit log of all detected items, classifications, timestamps, and reviewer IDs.
- FR9: System shall generate formatted compliance and monitoring reports.
- FR10: System shall support supervisor-driven updates to keyword lists and periodic retraining of classification models.

6. Non-Functional Requirements

- NFR1 (Availability): 24/7 operation, target uptime 99.5%+.
- NFR2 (Latency): Detection within seconds to minutes of broadcast or posting.
- NFR3 (Security): Read-only, role-based access, no write/delete permissions to source platforms or broadcast systems.
- NFR4 (Non-interference): no technical capability to alter, block, or remove monitored content.
- NFR5 (Auditability): All reports traceable, timestamped, and immutable.

- NFR6 (Language coverage): Urdu and English, including Roman-Urdu and code-switching.
- NFR7

7. Implementation Tiers [Cost, Engineering Effort, and Timeline]

Three implementation tiers are evaluated on technology/infrastructure cost only. Human staffing of 5–6 supervisors for the Moderate and Low-Cost tiers, or a larger reviewer team for the High Tier, is a constant organizational cost separate from this project analysis and is not included in each tier's cost table.

7.1 High Tier (Commercial Cloud APIs)

Fully managed, commercial-grade cloud services for every layer; the fastest and lowest-effort option to build, but the most expensive to run.

A legal roadblock for cloud-hosted APIs was a bill restricting cross border data transfer to international servers. No AWS or Azure region is physically located inside Pakistan; nearest regions are India (Mumbai/Pune) or the Middle East (UAE/Bahrain). The bill was defined as follows:

> *Pakistan's Personal Data Protection Bill (PDPB) 2023 establishes a restrictive framework for cross-border data transfers with significant localization requirements. The legislation categorizes personal data into three tiers: regular personal data, sensitive personal data, and critical personal data, with increasingly stringent transfer restrictions for each category. Critical personal data must be processed exclusively on servers within Pakistan, while sensitive personal data requires maintaining certain components locally according to mechanisms to be established by the proposed National Commission for Personal Data Protection.*

This bill was not passed but remains a real risk as consistent digital privacy reforms have been introduced in the past half decade, and data with national/provincial interest will inevitably

be revisited. This tier should be carefully considered and accepted only after substantial legal risk analysis.

Component	Estimated monthly cost
Speech-to-text (~14 channels, 336 channel-hrs/day, enterprise ASR API)	$1,000–4,000
OCR (on-screen text, commercial vision API)	$500–1,500
Text/video classification (commercial per-transaction APIs)	$1,500–5,000
Social platform API access (incl. Reddit's ~$12,000/yr enterprise floor; TikTok's ~$12,000/ floor)	$1,500–4,000
Cloud hosting & bandwidth (multi-region, high-capacity)	$3,000–8,000
Total estimated cost/month	**$7,500–22,500**

Architecture approach

Engineering effort. Low. Nearly all hard problems liketranscription accuracy, classification, scaling are solved by the vendor and the team primarily writes API integration code and webhook handlers.

Development time. ~4–8 weeks, including platform API approvals, which run in parallel with development and integration.

7.2 Low-Cost Self-Hosted Strategy (Fully Custom Build)

Open-source AI models, custom-built and self-managed end-to-end on owned GPU infrastructure. This option is the cheapest to run, but the most engineering-intensive to build.

Architecture approach

Audio layer. Self-hosted open-weight Whisper-family model (natively supports Urdu and English) on GPU infrastructure, with voice-activity-detection pre-filtering to reduce compute load.

Text layer. Self-hosted, open-source OCR (Urdu-trained) plus fine-tuned open-source Urdu/English/Roman-Urdu classification models, supplemented by a supervisor-maintained keyword engine.

Video layer. Open-source image classifier for periodic frame sampling, run on the same infrastructure.

Human layer. 5–6 supervisors handle keyword modification, classifier retraining, escalation triage, and report generation only.

*** One-time hardware investment (not included in monthly total below):** GPU/server hardware for self-hosted ASR and classification workloads utilizing Nvidia RTX A30; **$3,000–5,000, one-time CAPEX**.

Component	Estimated monthly cost
Server hosting, electricity, cooling	$300–600
OCR (self-hosted, open-source)	Negligible (~$0–50)
Text/video classification compute (self-hosted models)	$200–400
Social platform API access (Reddit/TikTok/IG/YouTube/Facebook free/low-volume tiers)	$0–200
Total estimated cost/month [excluding staff + API licenses]	**$500–1,250**

Engineering effort. High. Requires building a custom real-time streaming pipeline including audio ingestion, voice-activity detection, multi-channel scaling, fine-tuning classifiers on labeled data, and integrating a video/frame-sampling pipeline, but still requires a team owning uptime, scaling, and failure recovery. This also requires development to be onsite and integrated continuously with development decreasing feasibility.

Development time. ~4–8 months, dominated by data preparation, custom pipeline engineering, and iterative model tuning.

7.3 Moderate Tier (Self-Hosted on Pre-Built Deployment Software) [Recommended]

This tier keeps data and models on owned GPU hardware (CAPEX), like the Low-Cost tier, but replaces custom pipeline engineering with production-ready open-source deployment containers, cutting engineering effort significantly while keeping recurring cost low. This architecture requires additional contracted engineers for pipeline maintenance as it utilizes erratic open source container software.

Architecture approach

Speech-to-text. Deploy the official 'faster-whisper' Docker container, a ready-to-run transcription service with an OpenAI-compatible API and built-in streaming support. This involves configuration, not custom pipeline code.

OCR and classification. deploy NVIDIA Triton Inference Server's pre-built Docker container, which serves multiple models (OCR, text classifiers, video/image classifiers) from a single instance via configuration files, with built-in batching and multi-model management.

Hardware. A single higher capacity GPU (e.g., RTX A6000, 48GB VRAM) to run all model types concurrently without memoryoptimization engineering. This is a deliberate trade of for slightly higher CAPEX requiring less optimization and complex model tuning/testing.

Maintenance. A part-time contractor handles container updates, model swaps, and troubleshooting, or in-house IT staff (trained) as per the organization's resources/requirements.

***One-time hardware investment (not included in monthly total below):** GPU/server hardware (RTX A6000-class) $5,000–7,500, one-time CAPEX.

Component	Estimated monthly cost
Server hosting, electricity, cooling	$300–600
Social platform API access (free/low-volume tiers)	$0–200
Part-time DevOps/ML contractor retainer (container/model maintenance)	$1,000–2,000
Total estimated cost/month [excluding staff]	**$1,300–2,800**
Social platform commercial/enterprise API (Reddit, TikTok)	$2,000-$4,000

Total estimated cost/month [including API license)	**$3,300-$4,800**

Engineering effort. Low-to-moderate. The team configures and deploys well-documented, community-maintained serving software rather than building a pipeline from scratch but requires a contractor retainer who owns ongoing infrastructure upkeep.

Development time. ~3–6 months. Slightly faster than High Tiers speed, since deployment is largely configuration rather than custom engineering, while still retaining in-house self-hosted data control and trained staff.

7.4 Tier Comparison Summary

Tier	Monthly tech cost	One-time CAPEX	Engineering effort	Development time
High Tier (Commercial Cloud)	$22,000–61,500	None	Low	4–8 weeks
Low-Cost Self-Hosted (Custom Build)	$500–1,250	$3,000–5000	High	4–8 months
Moderate Tier (Recommended)	**$1,300–2,800**	**$5,000–7,500**	**Moderate**	**3-6 months**

The Moderate Tier is recommended because it captures the cost savings of full self-hosting relative to High Tier, while avoiding the time intensive custom-engineering timeline of the fully self-built option by relying on pre-built software instead of a fully cutom pipeline development.

8. Feasibility Assessment (Moderate & Low-Cost Tiers)

Economic feasibility

The Moderate Tier is cheaper on a recurring basis than the High Tier, with a moderate one-time hardware investment recovered quickly against equivalent hosted API spend. The Low-

Cost Tier is cheapest on a recurring basis, but at a substantially longer build timeline and engineering complexity.

Technical feasibility

Whisper-family models, NVIDIA Triton, and open-source Urdu NLP classifiers are performant enough to meet the bilingual, multi-format requirement without expensive vendor lock-in.

Operational feasibility

Capping the team at 5–6 supervisors/technical staff is achievable because their role is tuning the detection models rather than performing manual moderation as the Moderate Tier further reduces operational burden by delegating infrastructure upkeep to a part-time contractor.

Schedule feasibility

The Moderate Tier's timeline is comparable to the High Tier, unlike the Low-Cost Tier's 4–8 month build.

9. Risk Analysis

Development Risks

Risk 1: Underestimated data-preparation time

Across all self-hosted tiers, fine-tuning Urdu/Roman-Urdu classifiers requires labeled examples from pre-flagged content, and industry data consistently shows this phase eats the majority of project time, not including the model training itself, frequently causing 3–6 month builds to slip further when teams underestimate it upfront (Santhirasekaram, 2026).

Risk 2: GPU procurement and AI model integration difficulty (Moderate/Low-Cost tiers)

Self-hosting production AI workloads looks simple in a demo but becomes genuinely hard to sustain asit requires exact-matching GPU drivers and CUDA (NVIDIA GPU processing platform) versions that break easily on updates, careful memory management to run four different AI tasks like speech, OCR, text, and video classification simultaneously on shared hardware without slowdowns or crashes, and ongoing tuning of "batching" to balance speed against GPU efficiency. These are problems that only get worse as more channels are added. Even pre-built

tools like NVIDIA Triton, still require someone on the team to configure model setup, monitor performance, and fix failures when they occur. NVIDIA's own documentation and independent industry analyses agree that this demands dedicated GPU infrastructure expertise, and teams lacking it often find the hidden time and effort cost outweighs whatever they saved by avoiding commercial APIs. This is exactly why the Moderate Tier's part-time contractor retainer is not optional and is covering a real, specialized skill gap and losing access to that contractor would leave the in-house compliance team unable to keep the system reliably running (Yaari, 2025).

Risk 3: Platform API approval delays or rejection

Meta's Application Review can now take up to 20 days and a single rejection resets the queue, while Reddit's commercial-access approval has no published timeline at all (Czuryszkiewicz, 2026). If either stalls, the development schedule slips regardless of which cost tier is chosen. Reddit's free tier is rate-limited to ~100 queries/minutem so if monitoring volume grows or if Reddit audits and deems the use commercial, the organization should be prepared to move to the enterprise tier (~$12,000/year floor), as reflected in the High Tier model.

Furthermore, it is not possible to scan platform-wide TikTok content via an official API for active or domestic content moderation (Graham, 2026). While TikTok for Developers provides specialized data tools, they are strictly bounded by heavy access restrictions, data delays, and read-only limitations. You cannot use them to perform real-time automated scanning, automated censorship, or cross-platform content removal. Many compliance teams turn to custom web scrapers or unofficial third-party data APIs (such as Phyllo or EnsembleData) to aggregate public comments and trends. While these can extract public JSON data from video streams, they constantly fight severe rate-limiting blocks (HTTP 429 errors), require residential proxies to maintain, and technically violate TikTok's terms of service regarding automated scraping. Remaining within guidelines, APIs for TikTok continuous monitoring can range $500–$3,000/month through a third-party vendor, since TikTok itself offers no compliant free path for commercial content monitoring the way Instagram, YouTube, or Facebook do.

Operational Risks

Risk 4: Speech-recognition hallucination

Whisper-family models are not immune to errors and can generate fabricated sentences that were never spoken even during silence, background noise, or unclear speech. One peer-

reviewed study found hallucinations in roughly 1.4% of transcriptions, with 38% of those hallucinated segments containing harmful content (Koenecke et al., 2024). For a moderation and reporting system system, a hallucinated transcript could trigger a false flag or cause a real violation to be misclassified and dismissed.

Risk 5: False positives and false negatives in classification

This is the best-documented failure mode in AI content moderation. Automated systems routinely misclassify safe content as violations and miss genuine violations that lack the exact keywords or patterns the model was trained on. Research has documented that automated moderation systems disproportionately flag certain dialects and demographic speech patterns as violations even when content is almost identical to unflagged speech (L. Frank, 2026). Since the system covers Urdu, English, and Roman-Urdu simultaneously, uneven training data across these language creates an adverse output probability where regional dialects or code-switched speech may get flagged or missed at unpredictable rates than standard Urdu or English.

Risk 6: Model and keyword drift over time

Language evolves quickly on social media like new slang, coded terms, and evasion tactics emerging perpetually. Without ongoing retraining and keyword updates, which will be the core job of the indispensible 5–6 supervisors, detection accuracy degrades gradually and silently, and obvious violations may slip through.

Risk 7: Silent accuracy erosion from platform API changes

Instagram, TikTok, YouTube, and Facebook periodically change their API schemas, rate-limit formulas, and data fields without giving notice. A structural change could silently break part of ingestion pipeline as content stops flowing in, but the system doesn't always throw an exception or runtime error, so gaps in coverage may go undetected until a manual review identifies them.

Risk 8: Contractor/vendor dependency (Moderate Tier specifically)

The Moderate Tier's low engineering effort depends on a part-time DevOps/ML contractor for ongoing container and model maintenance. If that contractor becomes unavailable, in-house staff may lack the expertise to diagnose infrastructure issues quickly, creating a risk that doesn't exist in the fully in-house Low-Cost tier or the fully vendor-supported High Tier.

10. Feasibility Analysis

This project is achievable, but it is not simple to develop and integrate and it is not risk-free. Before moving forward, the organization should carefully consider the risk and feasibility analysis.

Project Difficulty

This is not a plug-and-play system. It requires continuous, real-time coverage of 14 TV channels and multiple social platforms, in two languages, across audio, text, and video all at once. Even the easier recommended option still needs real engineering work like setting up GPU servers, deploying AI models correctly, connecting five different platforms with five different rules, and training a small team to manage all of it. Nothing here is truly "set it and forget it." Every tier we looked at, even the fastest one, still takes weeks to months of real work before it runs reliably.

Contracted Development Complications

If the organization employs outside company or contractor to build this rather than delegating its in-house engineering teams, that creates new risks on top of the technical ones. A contractor may not fully understand the organization's specific compliance needs, may build something that is hard to hand over cleanly. The "Moderate Tier" plan that is recommended depends on a part-time outside contractor to keep the system running. Outsourcing saves effort up front but creates a dependency that must be actively managed.

Reddit and TikTok Moderation Exclusion

This is the most serious limitation and undercuts the objectives of the proposed system. The entire idea of this system is to replace slow manual checking with fast, automatic, always-on monitoring.

Reddit's free access is capped at a low request rate so if usage grows even slightly, Reddit may flag the traffic and block IPs, or require a costly commercial enterprise license to be purchased ($12,000/year).

TikTok is worse as its free monitoring option is officially not allowed for business use at all. There is no legal free way for a company to pull TikTok content automatically at scale. The only

path is paying a third-party company for access, which adds ongoing cost and a middleman the organization does not control.

In practice, this means the "automatic, continuous" objective of the system may quietly break down for two of the four platforms. If the organization cannot get full automatic access to Reddit and TikTok, someone will still need to check those platforms manually sometimes, which is exactly the manual, slow process this project was meant to eliminate. This does not make the project entirely unfeasible, but it means Instagram, YouTube, and Facebook may end up far better covered than Reddit and TikTok, and that gap needs to be visible and accepted upfront.

System Failure and Contingency Plan

Self-hosted systems that run on owned GPU hardware do not have the same safety net as big cloud companies. A single GPU server can fail, overheat, lose power or simply fail. When the system goes down, even for a few hours, nobody is watching those 14 channels and social feeds automatically during that time.

For these reasons the organization must have a manual backup plan ready before launch and should include:

- A basic UPS/battery backup so voltage fluctuations and short power cuts don't take the system offline immediately.
- A clear, written process for what the human supervisors do if the AI system goes down. For example, temporarily switching to manual spot-checks of the highest-risk channels and accounts until service is restored.
- Regular testing of the contingency plan.
- Automated alerts for system failures so gaps in coverage are short and identified quickly.

Conclusion

This system can be engineered, and the Moderate Tier remains the most balanced option in terms of cost, effort, and speed to build. However, it should be approved with the understanding that it takes real, ongoing effort to run, not just to build; hiring outside help reduces effort but creates a new dependency risk; Reddit and TikTok coverage will likely stay partial and unreliable, no matter what technical choice is made; and downtime will happen, so a tested manual contingency plan is a requirement for launch.

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